

Topic # 6A
Chapter 7 – PMBOK



Budget

- An estimate of income and expenditure for a set period of time.
- Predicts the future financial position of the project.
- There is not a single budget, but one for different levels.
- Budget helps to aid the planning of actual operations by forcing managers to consider how the conditions might change and what steps should be taken now and by encouraging managers to consider problems before they arise.
- Help to control resources and defines accountability

Ready made products v/s Make-to-Order

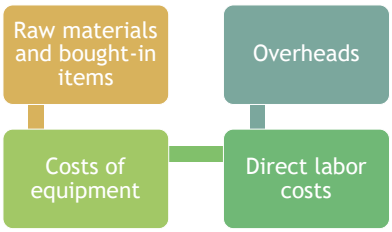
- Most S.W are developed according to specifications given by client.
- If the project is not too long, organization will most likely get paid once the project finishes. Until then it will have to operate on its resources.
- If it is a long term project, the payment may be made in parts.
- If the project will take around 2 years, it would mean recalculating the resources required for the project after a few months to check whether the project is on schedule and under budget.
- Sales versus order intake

Costing

- Usually a procedure to calculate the total amount spent and analyzed directly and indirectly, in producing or making the product.
- The price at which the company decides to sell its services or product depends upon the cost of producing or providing it.
- Cost depends upon factors such as
  - Price & availability of competing products
  - Demand
  - Availability of resources, actual cost, HR, technical, products utilized etc



Categories of Cost





Costs of Labor

- How do you quote the price for a software?
- Employee cost includes not just his salary, but other factors also weigh in.
- If there is a provident fund, social security, the organization will have to make contribution to it.
- Overtime & Employee may take leaves.
- Payroll/direct cost is the cost that is directly associated with employing a worker including benefits and salary
- Then there are other costs such as Insurance, overheads, such as electricity costs etc, these are known as labor burden.

- For detailed analysis, you can take into account time that the employee is expected to work.

TABLE 7.1 *Calculation of the number of revenue-earning hours in a year*

Total number of weekdays (1)	260
Public holidays (2)	10
Annual leave (3)	20
Sick leave (4)	5
Unproductive time (5)	10
Total non-revenue-earning time (2) + (3) + (4) + (5) = (6)	45
Total number of revenue earning days (1) – (5) = (7)	215
Total number of hours available (7) × 7	1505

- So we can find out direct cost of an employee per hour as: direct cost/ total number of hours avail.

Example

- Suppose the direct cost of technician is $22,000 so the cost of one hour would be 22,000/1505 =$ 14.62
- NOW if a cost of a basic component is $200 and it requires 10 hours of technician. Neglecting all other cost, What would be the total cost of this component?
- Component Cost + labor cost = 200 + 10*14.62 = $346.20

Overhead Costs



- Cannot be directly associated with a particular product
- Examples include rent, gas, electricity, and labor burden.
- Business ultimately has to pay for these costs so they have to be added to the direct costs of the product by one way or other.
- One way of allocating overhead costs to the product cost…. Simplest way
- By setting a fixed percentage on total direct costs or labor costs
- For example overhead costs are set to 30% of direct labor costs
- So if the direct labor costs amounts to $146.2 therefore the overhead will be 30% of $146.2

- Another way is to distribute the overhead over multiple products
- Example Suppose total overhead amount to be $63,500 yearly and company has following yearly sales forecast producing three models of computer. Technician costs $14.62 per hour

- Allocate same overhead costs irrespective of direct costs associated.
- Total sales in units = 200+100+50 = 350
- Overhead cost per unit= 63,500/350= 181.43
- ***Cost of basic model = Component cost + labor cost + overhead cost***
- Basic model cost = $200+ 10*14.62 + 181.43
- Total cost = 527.63

TABLE 7.2 *Direct costs and expected sales for the different computer models*

Model	Cost of components ($)	Technician time (hours	Expected sales
Basic	$200	10 hours	200
Advanced	$300	12 hours	100
Professional	$400	15 hours	50

- Second way is to divide the over head per labor hours
- Solution:
  - 3 technicians involved, each yearly for 1505 hours. Hence total labor hour would be 4515 labor hours
  - Overhead cost per labor hour = 63500/4515= $14.06 per hour
  - ***Basic model will cost = component + labor + overhead***
  - 200 + (10*14.62) +(10*14.06)= 486.8



Project Cost Management

- Cost is a resource sacrificed or foregone to achieve a specific objective.
- ***Estimating costs*** involves developing an approximation or estimate of the costs of the resources needed to complete a project.
  - O/P:
    - cost estimates, basis of estimates & project document updates.
- ***Determining the budget*** involves allocating the overall cost estimate to individual work items to establish a baseline for measuring performance.
  - O/P:
    - cost performance baseline, project funding requirements & project document updates.
- ***Controlling costs*** involves controlling changes to the project budget.
  - O/P:
    - Work performance measurements, budget forecasts, organizational process asset updates, change requests, project management plan updates & project document updates.

Reserves
- Are dollars included in a cost estimate to mitigate cost risk by allowing for future situations that are difficult to predict.

Contingency Reserves (known unknowns)
- Allow for future situations that may be partially planned for and are included in the project cost baseline.

Management Reserves (unknown unknowns)
- Allow for future situations that are unpredictable



Resource Planning

Resource Planning

- To estimate budgets and control costs, project managers & their teams must determine what physical resources & what quantities of those resources are required to complete the project.
- Important points to ponder for resource planning:
  - Level of difficulty of a task
  - Uniqueness in project scope
  - Org history in doing similar tasks.
  - Does the org have physical resources that are capable & available for performing the work?
  - Org policies that might affect the availability of the resources.



Type of Estimate	When Done	Why Done	How Accurate
Rough Order of Magnitude (ROM)	Very early in the project life cycle, often 3–5 years before project completion	Provides estimate of cost for selection decisions	–50% to +100%
Budgetary	Early, 1–2 years out	Puts dollars in the budget plans	–10% to +25%
Definitive	Later in the project, less than 1 year out	Provides details for purchases, estimates actual costs	–5% to +10%

- Rough order of magnitude is also referred to as a ballpark estimate, a guesstimate, a swag, or a broad gauge.
- Definitive estimates are used for making many purchasing decisions for which accurate estimates are required and for estimating final project costs.
- It is important to provide supporting details (assumptions, project scope, WBS, etc) used in computing estimates so that it will be easier to prepare updates as needed or similar estimates on other projects.
- A **cost management plan** is a document that describes how the organization will manage cost variance on the project
  - A large percentage of total project costs are often labor costs, so project managers must develop and track estimates for labor

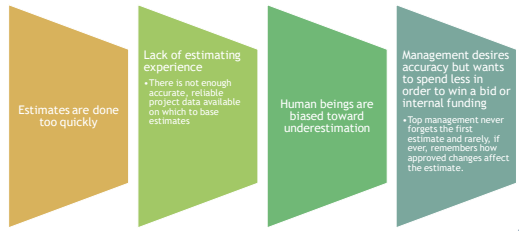


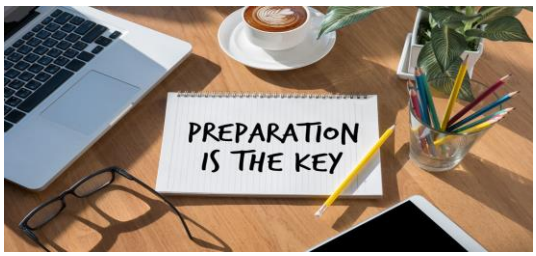
Cost/Budget Estimate tools & Techniques

- ***Top-down estimates/budgeting*** : It uses the actual cost of a previous, similar project as the basis for estimating the cost of the current project
  - Less costly but less accurate
  - How similar the current and previous project are determines the accuracy of the estimate. Using a different language or hardware can skew the estimate
- ***Bottom-up estimates/ budgets*** : It involves estimating individual work items or activities and summing them to get a project total.
  - The smaller the work items, the better the estimate.
  - These estimates are usually time intensive and expensive to develop
  - If a detailed WBS is available for a project, the project manager could have each person responsible for a work package develop his or her own cost estimate for that work package, or at least an estimate of the amount of resources required.

- ***Parametric modeling:*** uses project characteristics (parameters) in a mathematical model to estimate project costs
  - For example, a model might provide an estimate of $50 per line of code for a s/w development project based on the programming language, level of expertise of the programmers, size and complexity of the data involved, etc.
  - Some models may be simpler such as a $10,000 ballpark estimate per workstation in a large office automation project based on history of similar projects during the same time period
- One popular parametric model is the ***Constructive Cost Model (COCOMO)*** which is used to estimate software development costs based on parameters such as the source lines of codes or function points.
  - ***Function Points*** are technology-independent assessments of the functions involved in developing the systems.
  - **Number of inputs, outputs, queries, file stored, external files interacted etc will determine the complexity of software and eventually the function points**

IT Projects Cost Estimates Problems





Preparing Cost Estimates

- Before creating an estimate, know what it will be used for, gather as much information as possible, and clarify the ground rules and assumptions for the estimate
- If possible, estimate costs by major WBS categories
- Create a cost model to make it easy to make changes to and document the estimate

- ***EXAMPLE OF COST ESTIMATION CREATION.***
  - Working hours in a month is 160 hours.
  - A project manager works part time on this project for 12 months and costs $100/hour.
  - Four additional team members required that give 25% of their time each on the project for 12 months and cost $75/hour for project management tasks. Contracted labor estimate costs are 10% of total software development & testing costs.
  - Handheld devices: 100 devices @ $600 per unit.
  - Servers: 4 servers estimated @ $4,000 each, based on recent server purchases.
  - Licensed S/w: A cost of $200/handheld device will be used.
  - Software Development: Labor approach is used. Labor hours & cost Given by the supplier who did the development.
  - Testing: Based on similar projects, testing will be estimated as 10 percent of the total hardware and software cost.

Example Continued

- **Training and support**:
  - Based on similar projects, training will be estimated on a per-trainee basis, plus travel costs.
  - The cost per trainee (100 total) will be $500, and travel will be $700/day for the instructors and project team members. It is estimated that there will be a total of 12 travel days.
  - Labor costs for the project team members will be added to this estimate to assist in training and providing support after the training.
  - The same team members again spend 25% of time each on training. The labor hours estimate for team members is 1,920 hours total.
- ***Reserves***:
  - As directed, reserves will be estimated at 20 percent of the total estimate.

	# Units/Hrs.	Cost/Unit/Hr.	Subtotals	WBS Level 1 Totals	% of Total
WBS Items					
1. Project Management				**$306,300**	**20%**
Project manager	960	$100	$96,000		
Project team members	1920	$75	$144,000		
Contractors (10% of software development and testing)			$66,300		
2. Hardware				**$76,000**	**5%**
2.1 Handheld devices	100	$600	$60,000		
2.2 Servers	4	$4,000	$16,000		
3. Software				**$614,000**	**40%**
3.1 Licensed software	100	$200	$20,000		
3.2 Software development*			$594,000		
4. Testing (10% of total hardware and software costs)			$69,000	**$69,000**	**5%**
5. Training and Support				**$202,400**	**13%**
Trainee cost	100	$500	$50,000		
Travel cost	12	$700	$8,400		
Project team members	1920	$75	$144,000		
6. Reserves (20% of total estimate)			$253,540	**$253,540**	**17%**
Total project cost estimate				**$1,521,240**	

* See software development estimate

Software Development Cost

1. Labor Estimate	# Units/Hrs.	Cost/Unit/Hr.	Subtotals	Calculations
Contractor labor estimate	3000	$150	$450,000	3000*150
Project team member estimate	1920	$75	$144,000	1920*75
Total labor estimate			**$594,000**	Sum above two values



Cost Budgeting

- ***Cost budgeting*** involves allocating the project cost estimate to individual work items over time
- The ***WBS*** is a required input to the cost budgeting process since it defines the work items
- An important goal is to produce a ***cost baseline***
- A ***time-phased budget*** that project managers use to measure and monitor cost performance
- Estimating costs for each major project activity over time provides management with a foundation for project cost control
- Cost budgeting also provides info for project funding requirements –at what point(s) in time will the money be needed

Example

- Project Manager & Team Members work 12 months equally
- Contractors start working from 2nd month and work till end equally for 11 months
- Half purchases of handheld devices & servers are done on 4th month & half on 5th month
- Licensed software is purchased along with the purchase of handheld devices.
- Development starts on 2nd month & ends before training
- Testing starts on 3rd month and ends before training
- Training costs incurred in the 9th month
- Rest are all assumptions

Budget Within Time

WBS Items	1	2	3	4	5	6	7	8	9	10	11	12	Totals
1. Project Management													
Project manager	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	8,000	96,000
Project team members	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	12,000	144,000
Contractors		6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	6,027	66,300
2. Hardware													
2.1 Handheld devices				30,000	30,000								60,000
2.2 Servers				8,000	8,000								16,000
3. Software													
3.1 Licensed software				10,000	10,000								20,000
3.2 Software development		60,000	60,000	80,000	127,000	127,000	90,000	50,000					594,000
4. Testing			6,000	8,000	12,000	15,000	15,000	13,000					69,000
5. Training and Support													
Trainee cost									50,000				50,000
Travel cost									8,400				8,400
Project team members							24,000	24,000	24,000	24,000	24,000	24,000	144,000
6. Reserves				10,000	10,000	30,000	30,000	60,000	40,000	40,000	30,000	3,540	253,540
Totals	20,000	86,027	92,027	172,027	223,027	198,027	185,027	173,027	148,427	90,027	80,027	53,567	1,521,240

END OF TOPIC 6A

- **COMING UP!!!!!!**
- **Project Cost Management (contd..)**
- **Midterm Exam II**
- **Resource Allocation**
- **Project Control**
- **Agile Software Project Management**